

Failure of Answer-only Evaluation

Q: How many coconuts are there in the food boat?



Model answer: 9 ✓ (Correct)

Predicted Evidence Mask:



Text Accuracy: 1

Considered correct (but grounding is wrong)

Our Benchmark: Joint Evaluation

VISTAQA :



task types:

Attribute, OCR, Identification, Counting, Comparison, Spatial



domains:

Indoor, Outdoor, Math, Science, Robotics, Autonomous Driving



1157 Expert-curated samples

Joint Metric: GROVE

No compensation across modalities

Text accuracy : 1

Mask Quality (mIoU): 0.22

GROVE = 0.47 ✗ (Low)

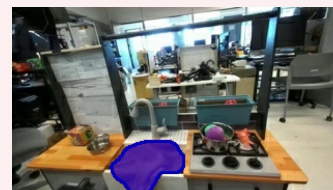
Failure of Grounding-only Evaluation

Q: What color is the object that the robotic arm would need to move to clean the sink?



Model answer: blue ✗ (wrong)

Predicted Evidence Mask:



IoU: 0.95

Considered good (but answer is wrong)

Attribute

AV



Q: What driving behavior should the ego vehicle adopt based on the sign on the left?

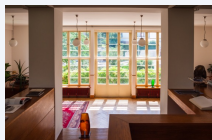
A: reducing speed.

GT Mask:



Identification

Indoor



Q: Which objects are used to control indoor temperature?

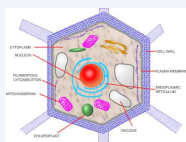
A: radiators.

GT Mask:



OCR

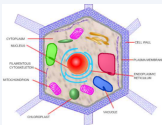
Science



Q: What is the name of the white structures shown in the cell?

A: Vacuole.

GT Mask:



Counting

Indoor



Q: How many black bags are there in this image?

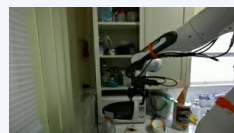
A: 10.

GT Mask:



Comparison

Robotics



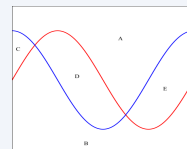
Q: Between the toaster and the orange kettle, which object is positioned closer to the robotic arm's gripper?

A: There is no toaster or orange kettle visible in the image.

GT Mask: None (hallucination)

Spatial

Math



Q: Which regions are enclosed between $\sin(x)$ and $\cos(x)$ where $\cos(x) < \sin(x)$?

A: Region D.

GT Mask:

